

ROVER LAB, DOSSIER (GJ 1214 b mission)

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Planet

Name: GJ 1214 b

Planet type: Warm sub-Neptune ('steam / haze world'), no solid surface

Surface temperature: ≈ 280 °C dayside / ≈ 164 °C nightside (both JWST-measured)

Surface pressure: No solid surface; deep, high-pressure atmosphere

Gravity: ≈ 1.13 g

Atmosphere: Thick, high-altitude haze/clouds (measured, high Bond albedo ~ 0.51); likely a metal-rich, high-metallicity atmosphere with tentative (unconfirmed) hints of water vapour and CO₂ beneath the haze

Mission Briefing

GJ 1214 b is a warm, cloud-choked world where a thick haze hides the atmosphere's true composition from orbit, one of the most-studied 'mystery' planets of its kind. A probe, 'Haze-runner', has punched through the cloud deck and returned with a sample of orange-tinted condensate from deep within the haze layer. Given how metal-rich this atmosphere appears to be, mission control wonders whether the droplets could actually be a concentrated iron-bearing brine, but the thick clouds have made it impossible to confirm anything remotely, so the sample will have to speak for itself.

Rover

Starting energy: 100

Site visit cost: 10

Experiment cost: 15

Overview Image

Site: Orange condensate from deep within the haze layer

Description: A vial of orange, slightly cloudy liquid collected from deep inside the planet's dense, swirling haze layer.

Identity: Food-colourant solution (orange dye in water), no real iron content. DECOY GROUP.

Known Properties

Colour: Orange, slightly turbid

Odour: None

Solubility: Fully miscible with water

Experiment Results

Measure pH: Close to neutral, pH around 6.5–7

Measure density: About 1.0 g/mL

Test electrical conductivity: Low

Add a magnet: No attraction

Add potassium thiocyanate solution: No colour change

Add sodium hydroxide solution: No precipitate forms

Heat a small sample gently: No visible change

Experiments

Measure pH

Measure density

Test electrical conductivity

Add a magnet

Add potassium thiocyanate solution

Add sodium hydroxide solution

Heat a small sample gently

AI Behaviour

Reveal remaining candidates after narrowing: No

Socratic questions: Yes

Difficulty: Standard (15-16 years old)

Teacher Notes

DECOY GROUP, assign a food-colourant solution, not real FeCl_3 . GJ 1214b's genuinely metal-rich, hazy atmosphere makes 'maybe it's iron-rich' a plausible-sounding guess, which is exactly the trap: the planet has no solid surface at all, and its real atmospheric chemistry below the haze is still an open scientific question, nobody has actually confirmed iron chloride there.

Real-science notes: GJ 1214b is a genuine, much-studied sub-Neptune, ~ 48 light-years away, discovered in 2009, orbiting its red dwarf star in just 1.58 days. Pre-JWST spectra were famously flat/featureless, which was itself informative (ruled out a clear, low-metallicity H/He atmosphere). JWST (Kempton et al. 2023, Nature) directly measured a high Bond albedo (~ 0.51) and large day-night temperature swing, both real measurements pointing to reflective high-altitude clouds/haze and a high-metallicity atmosphere. Later JWST work (2024–2025) found tentative (not confirmed) hints of water vapour and CO_2 beneath the haze. Whether the planet has a genuine deep water ocean or is closer to a metal-rich 'super-Venus' with no ocean at all is still actively debated, present as an open question if students research it.

Sources: Kempton et al. 2023, Nature ([nature.com/articles/s41586-023-05951-7](https://www.nature.com/articles/s41586-023-05951-7)); follow-up 2025 (iopscience.iop.org/article/10.3847/2041-8213/ada02c).

Cross-subject link: Biology selective pressure for this planet, crushing atmospheric pressure and a permanent haze that blocks almost all sunlight from reaching lower layers.